Welcome to

Fundamentals of Big Data Analytics

Lecture 1- Introduction

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Customize

FBDA- Spring 2023

Section- 4A

Class code

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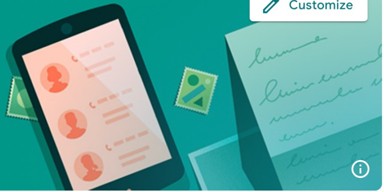
Section-4B



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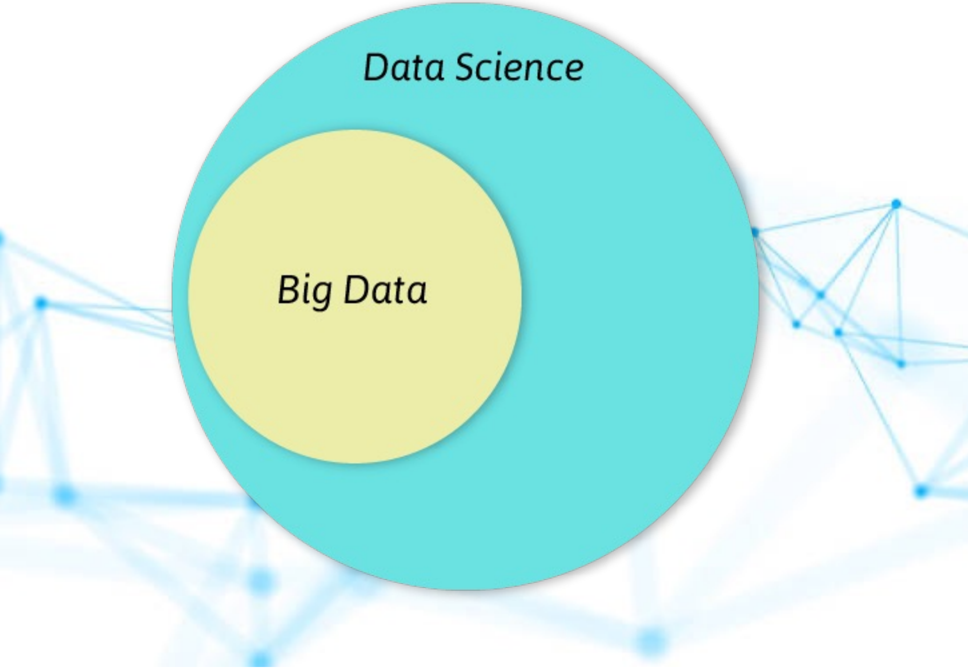
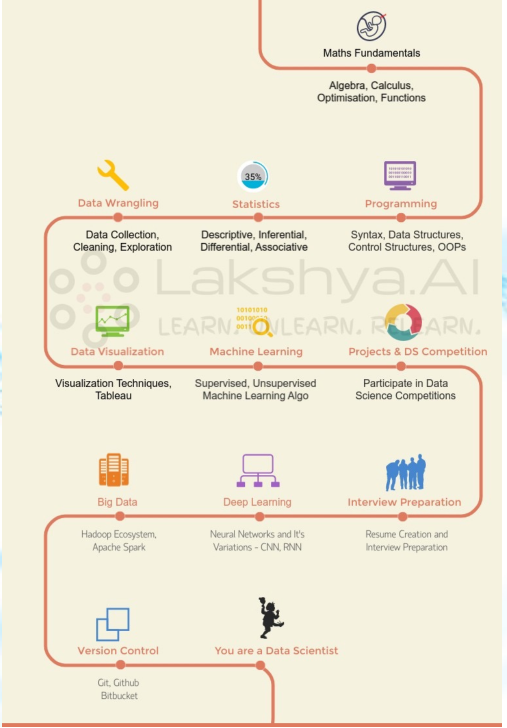

Data Science
Big Data


Maths Fundamentals
Algebra, Calculus, Optimisation, Functions
Data Wrangling
Data Collection, Cleaning, Exploration
Statistics
Descriptive, Inferential, Differential, Associative
Programming
Syntax, Data Structures, Control Structures, OOPs
Data Visualization
Visualization Techniques, Tableau
Machine Learning
Supervised, Unsupervised Machine Learning Algo
Projects & DS Competition
Participate in Data Science Competitions
Big Data
Hadoop Ecosystem, Apache Spark
Deep Learning
Neural Networks and It's Variations - CNN, RNN
Interview Preparation
Resume Creation and Interview Preparation
Version Control
Git, Github Bitbucket
You are a Data Scientist

Introduction

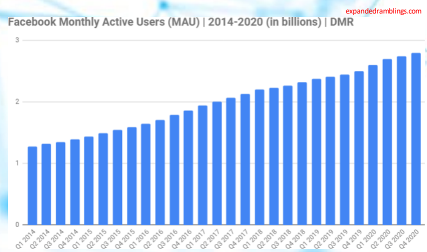
Big Data Analytics

- Big Data Generation and Growth
- What is Big Data
- Importance of Big Data Analytics
- Industries benefiting from Data Analytics
- Sources of Data (people, machines, organizations)
- Aspects of Bigness (The 5 V's of big data)
- Types of Data (table, text, multimedia, stream, sequence, graphs)
- The Analytics Process (preprocessing, analytics, visualization)

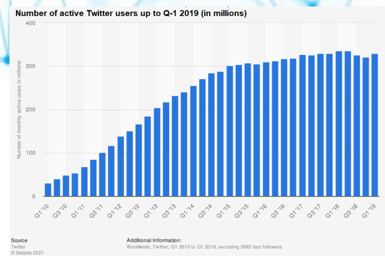
Big Data Generation and Growth

- Data has been generated at an exploding rate in recent years
- Organizations collect trillions of bytes of information about their customers, suppliers, and operations every day
- Large pools of data is being captured, communicated, aggregated, stored, and analyzed by businesses, academia, and governments
- Individuals with smartphones on social network sites are continuously fueling the exponential growth of multimedia data

Big Data Generation and Growth



https://expandedramblings.com/



Big Data Generation and Growth

Where data comes from?

- Internet users generate about 2.5 quintillion bytes of data each day[1]
- In 2018, internet users spent 2.8 million years online[2]
  - Social media accounts for 33% of the total time spent online [2]
- In 2019, there were 2.3 billion active Facebook users
- Twitter users send nearly half a million tweets every minute[1]
- By 2020, every person will generate 1.7 megabytes in just a second[1]

By 2020, there will be 40 trillion gigabytes of data (40 zettabytes)[3]

90% of all data has been created in the last two years [4]

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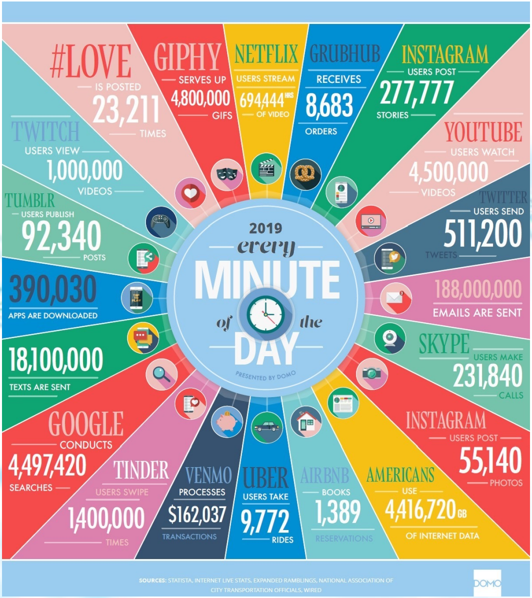
1 Domo report (a company with data analytic platform for businesses)

2 Global Web Index report (a company with big data analytic platform)

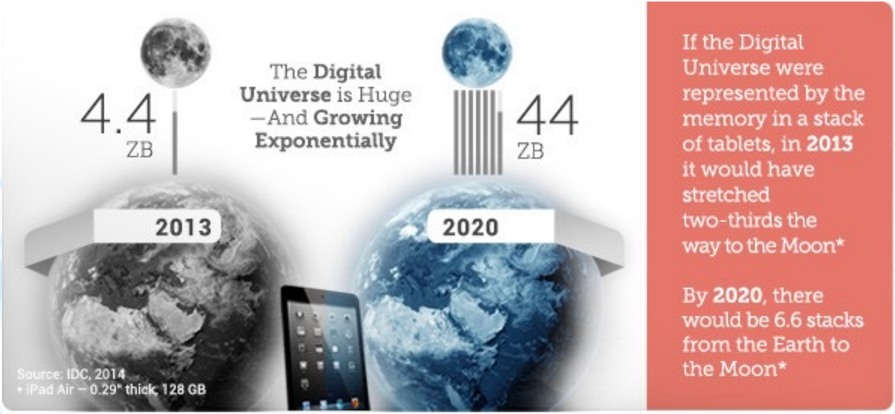
3 EMC (Dell EMC provides big data solutions)

4 IBM

Big Data Generation and Growth



Big Data Generation and Growth



- 90% of all data has been created in the last two years [5]

[5] IBM

What is Big Data

- **"Big data"**: datasets whose size is beyond the ability of typical database software tools to capture, store, manage, and analyze

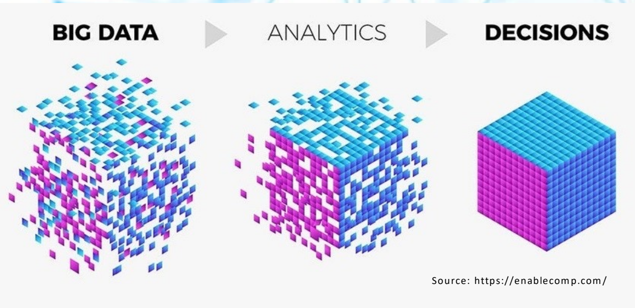
- As technology advances over time, the size of datasets that qualify as big data will also increase

- The definition varies by sector, depending on the kinds of available software tools and sizes of datasets in a particular industry

- With those caveats, big data in many sectors today will range from a few dozen terabytes to multiple petabytes (thousands of terabytes)

Data Analytics

- Data:Set of values of qualitative or quantitative variables
- Information:Meaningful or organized data
- Data Analytics:The process of examining data in order to draw and communicate useful conclusions about the information it contains



Analysis vs. Analytics

Analysis: Making use of past data and deriving results from the data

Analytics: Using data to obtain future insights and details regarding trends etc..

Big Data Vs Data Science

Factors	Big Data	Data Science
Concept	Handling large data	Analyzing data
Responsibility	Process huge volumes of data and generate insights	Understand pattern within data and make decisions
Industry	E-commerce, security services, telecommunication	Sales, image recognition, advertisement, risk analytics
Tools	Hadoop, Spark, Flink	SAS, R, Python

Big Data Vs Big Data Analytics Vs Data Science: Tabular Comparison

Basis	Big Data	Big Data Analytics	Data Science
Definition	Data generated from variety of sources in huge volume with high velocity is Big Data	Use of specialized software and tools to analyze big data to conclude decision making	Use of multi disciplines to extract information and interpretable insights from structured and unstructured data
Skill sets	Set up/Maintain Infrastructure Data preprocessing API development Data pipeline implementation and monitoring	Analyze Data Data Modeling Data Engineering Storyboarding Business Analytics Business Intelligence	Machine Learning Exploratory Analysis Data Mining Modeling Visualisation Software Development
Application Areas	Financial Services, Retail, Health & Sports, performance optimization, Communication, etc.	Healthcare, Travel, Gaming, Energy Management, etc.	Internet Searches, Search Recommenders, Digital Advertisements, Image/Speech recognition, Fraud & risk detection, etc.
Tools	Kafka & Storm Cloud Computing AWS/Azure/GCP	KNIME, SAS, Python Power BI/Tableau	Apache Spark/Hive Neural Networks frameworks R or Python

Course Details

Concepts and practical techniques for Big Data Analytics

What are Big Data frames works and tools

-How to analyze data to get data insights

-Data preprocessing and transformation

- Machine learning for Text classification/ Clustering

-Data visualization tools and techniques

Course Prerequisites and Format

Prerequisites:

-Basic concepts of computer science (e.g. data structures)
-Comfortable with programming, particularly with C++
-(Optional) Knowledge of Python, Data Science and Data Ba
Systems

-**Format**:

-Lecture + quizzes + assignments + term project
+ mid term + final exam

Grading Scheme: Absolute

Course Book

Mining of Massive Datasets

By

Jure Leskovec Stanford Univ. Anand Rajaraman Milliway Labs , Jeffrey D. Ullman Stanford Univ.

Big Data Analytics: Market

Data Analytics: Then and Now

- Data Analytics has been around for years
- Even in 1950's, businesses were using basic analytics (manual examination) on data (essentially numbers in a spreadsheet) to uncover insights and trends
- New tools and technologies bring speed and efficiency in techniques
- Today, businesses analyze data and can identify insights for immediate decisions
- The ability to work faster and stay agile gives organizations a competitive edge they did not have before

Why is Big Data Analytics Important

Organizations analyze data

- to identify new opportunities
- to gain insights that lead to smarter business decisions to
- identify methods for more efficient operations
- to maximize larger revenues and higher profits to
- keeps customers satisfied

Top three factors businesses got the most value in

- Cost reduction
- Faster, better decision making
- New products and services

Why enterprises use Big Data Analytics

Companies are using big data analytics for all types of decisions


The Evolution of Decision Making:
How Leading Organizations Are
Adopting a Data-Driven Culture

A REPORT BY **HARVARD BUSINESS REVIEW ANALYTIC SERVICES**

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sas THE POWER TO KNOW.

Harvard Business Review

What enterprises use Big Data Analytics for

- Competitor Analysis
  - Online traffic to websites and related social media
- Market Analysis
  - Trends and market segment analysis
- Productivity Enhancement
  - Analyze employees tracking data
- Cost Cutting
  - Reduce energy bills, optimize routes, predict demands, process efficiency and automation[6]
- Targeted Marketing
  - Analyze purchasing history and target the right people for a product
- Improved Customer Relation

  Analyze customer feedback and make adjustments

[6] Forbes (01/08/2016) Big Data Analytics' Potential to Revolutionize Manufacturing Is Within Reach

Industries Benefiting from Big Data Analytics

- Retail: Advertising, Targeted marketing, recommendation system, customer loyalty, inventory management, demand prediction
- Banking and Financial: Customer loyalty and churn, fraud detection, risk assessment
- Brands: 66% brands use data analytics for product and service launch, appropriate timings
- Logistics and Transportation: Fleet management, maintenance needs, drivers risk assessment, real time tracking
- Health Care: Efficiency in healthcare operations, predictive analytics, outbreak prediction, immunization strategy

Google's AI system can beat doctors at detecting breast cancer

By Hanna Ziady, CNN BUSINESS January 2, 2020

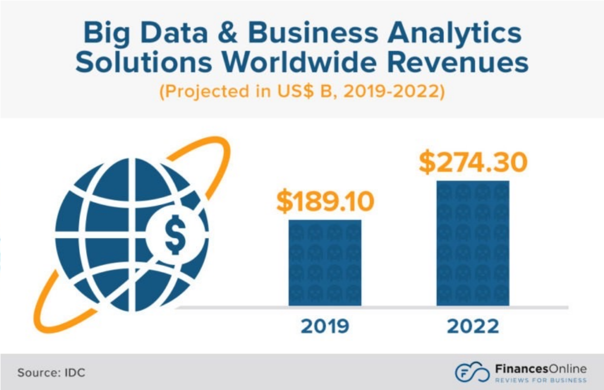
- Government & Utility Companies: Surveys & census, development planning, health, education, energy supply & demand management

Big Data Analytics - Market

- 12% - the rate of increase for big data and business analytics use from 2018 to 2019 [7]
- \$189.1 billion – projected worldwide revenues for big data and business analytics solutions for 2019 [7]
- \$274.3 billion – projected worldwide revenues for big data and business analytics solutions by 2022 [7]
- 13.2% - projected compound annual growth rate (CAGR) of big data and business analytics within the five-year period, 2018-2022 [7]

[7] International Data Corporation (IDC) - Big data analytics company

Big Data Analytics - Market



Sources of Big Data

Sources of Big Data

ELSEVIER

Information Processing & Management
Volume 54, Issue 5, September 2018, Pages 758-790

A survey towards an integration of big data analytics to big insights for value-creation

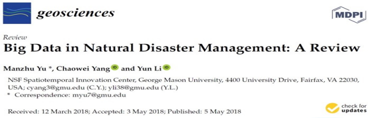
Mandeep Kaur Saggi, Sushma Jain

Sources: Machine Generated Data

- Biggest source of big data
- Temperature sensors, GPS navigator, Satellite imagery,
- Apps, Increasing number of smart devices, IoT
- A 12 hours flight produces 84TB of data, sensors, temperature, pressure, accelerometer, turbulence
- Smart City, Smart Transportation
- Think about the volume of video data collected at Lahore Safe City Authority Control Room
- Generally, such data is unstructured

Sources: People Generated Data

- Blogs, social network posts, keywords search, photo sharing, pictures, emails, ratings and reviews
- Daily facebook data 30+ PB > All US Academic libraries (2 PB)
- Companies use 12PB/day Twitter data for sentiment analysis around their products
- Could be used for disaster management, e.g. to identify and measure affected areas and channel resources


geosciences MDPI

Review

Big Data in Natural Disaster Management: A Review

Manzhu Yu *, Chaowei Yang and Yun Li

NSF Spatiotemporal Innovation Center, George Mason University, 4400 University Drive, Fairfax, VA 22030, USA; cyang3@gmu.edu (C.Y.); yli38@gmu.edu (Y.L.)
* Correspondence: myu7@gmu.edu

Received: 12 March 2018; Accepted: 3 May 2018; Published: 5 May 2018

check for updates

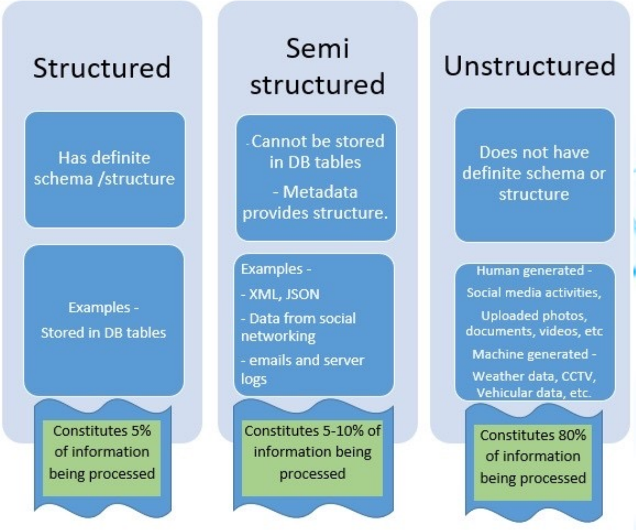
- Typically unstructured, or at best semi-structured such as emails, where the header has somewhat of a structure, except in few cases such as filling up a survey form
- Generally more text: 500 million tweets per day

Sources: Organization Generated Data

- FAST Students Data, ESPN Cricinfo, TCS shipment tracking data
- Governments open data, Stock Records, Banks, e-
- Commerce Medical Records
- Optimize routs and optimal scheduling can save 50m by reducing each drivers route by one mile
- Combine Walmart sales data with Twitter sentiment analyses or events to launch a new product
- Estimate
- demands Fraud
- Detection

Highly Structured Data

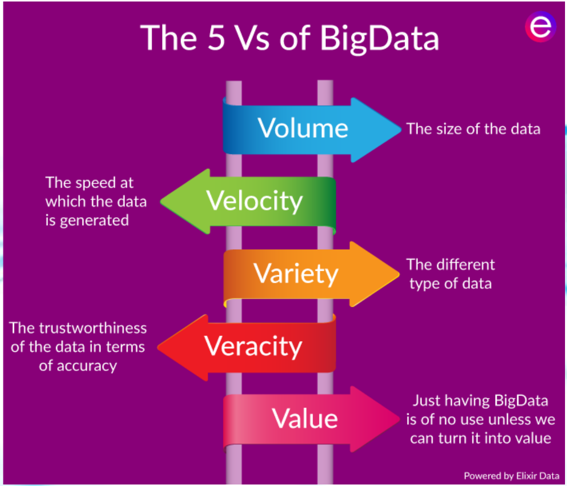
Categories of Data



The 5 V's of Data

Aspects of Big: The 5 V's

1. Volume
2. Velocity
3. Variety
4. Veracity
5. Value



Aspects of Big: The 5 V's – Volume

Volume: size, scale, dimensionality,

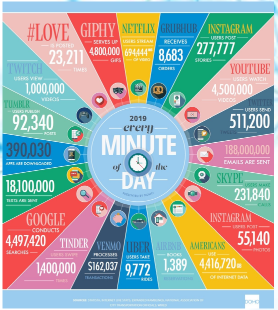
- 204m emails/minute, if an email is 100KB, see the volume

- Challenges: Acquisition, Storage, Retrieval, Processing Time
- Large dimensional data has more information, it is a blessing
- It is a also a big curse, dealing with large dimensions is a core topic in this course

Aspects of Big: The 5 V's – Velocity

Velocity: Speed of data is very high

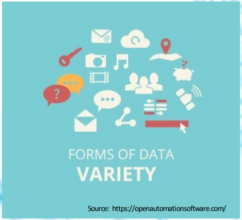
- Number of emails, twitter messages, photos, videos etc. per second



- Late decisions implies missed opportunities
- Real time processing vs Batch Processing (end of the day)

Aspects of Big: The 5 V's – Variety

Variety: Structural variety, different formats, models



- Medium variety, audio, text, video,
- DBMS, files, traffic logs, XML, code
- Online vs Offline,
- Real time vs Intermittent data (another way data varies)
- Challenges: requirement of analytics, Semantic, how to interpret

Aspects of Big: The 5 V's – Veracity

Veracity: Quality of data

- Data could have many issues (biases, anomalies, inconsistent measurements and units, incomplete and duplicate records)
- Volatility in data, updated/outdated, changing trends/sentiments
- Trustworthiness and reliability of sources and generation/processing
- Fake news, rumours, fake likes, fake followers

Aspects of Big: The 5 V's – Value

Value: Data can be turned into big value

- Data having no value is of no good to the company
- Should be able to meet strategic objectives
- Should amplify other technology innovations

5 Vs of Big Data: Value

The Economist Intelligence Unit report on surveying 476 executives

- 60% feel that data is generating revenue within their organizations
- 83% say it is making existing services and products more profitable
- 63% executives based in Asia said they are routinely generating value from data
- In the US, the figure was 58% and in Europe, 56%


35,524 views | Jan 13, 2016, 02:24am

Big Data Facts: How Many Companies Are Really Making Money From Their Data?

Bernard Marr Contributor
Enterprise Tech

Forbes

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